Data Sciences – CentraleSupélec
Advanced Machine Learning
Course II – Linear Regression and Regularization

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Linear regression

Motivations:

- Simple and interpretable baseline, essential for understanding more sophisticated models.
- Explicit description of the relationships between inputs and outputs.
- Can outperform more flexible models when training data are scarce, noisy, or high-dimensional.
- Extended applicability when combined with nonlinear basis functions (see Lab).

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Applications: Prediction of

- sales or economic indicators from explanatory variables;
- house prices from property characteristics;
- quantitative outcomes from behavioural or biomedical measurements.

Linear model

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The inputs may also be transformed features or basis expansions.

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$$y_i \approx f_{\boldsymbol{\beta}}(\mathbf{x}_i) = \beta_0 + \sum_{j=1}^{d} \beta_j x_{ij}.$$

To include an intercept, we augment each feature vector with a constant 1. In the following, $\mathbf{x}_i$ denotes this augmented vector.

$$\mathbf{x}_i = [1, x_{i1}, \ldots, x_{id}]^\top, \qquad \boldsymbol{\beta} = [\beta_0, \beta_1, \ldots, \beta_d]^\top,$$

we obtain

$$f_{\boldsymbol{\beta}}(\mathbf{x}_i) = \mathbf{x}_i^\top \boldsymbol{\beta} = [\mathbf{X}\boldsymbol{\beta}]_i,$$

where $\mathbf{X} \in \mathbb{R}^{n \times (d+1)}$ contains $\mathbf{x}_i^\top$ as its i-th row.

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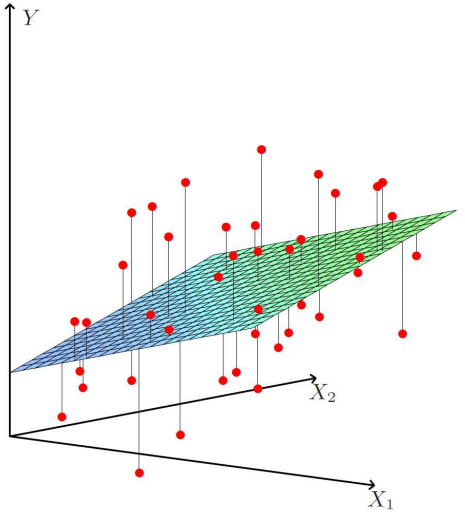
Goal: Learn $\boldsymbol{\beta}$ from the training data in order to predict a real-valued output y for a new observation $\mathbf{x}$.

Least Squares

Principle: Find β minimizing the sum of squared residuals:

$$F(\beta) = \frac{1}{2}\sum_{i=1}^{n}(y_i - f_\beta(\mathbf{x}_i))^2 = \frac{1}{2}\|\mathbf{X}\beta - \mathbf{y}\|^2 = \frac{1}{2}\|\mathbf{e}\|^2$$

with $\mathbf{e} = \mathbf{y} - \mathbf{X}\beta$ the residual vector.



Least squares solution

The least-squares criterion is

$$F(\boldsymbol{\beta}) = \frac{1}{2}\|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|^2.$$

Its gradient reads

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If $\mathbf{X}$ has full column rank, $\mathbf{X}^\top\mathbf{X}$ is positive definite and the solution is unique:

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}.}$$

Geometrical interpretation

The fitted values at the training inputs are

$$\widehat{\mathbf{y}} = \mathbf{X}\widehat{\boldsymbol{\beta}} = \underbrace{\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top}_{\mathbf{H}}\mathbf{y},$$

where

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is the **hat matrix**.
$\mathbf{H}$ is the orthogonal projector onto the column space $\text{col}(\mathbf{X})$.
Hence

$$\boxed{\widehat{\mathbf{y}} = \text{proj}_{\text{col}(\mathbf{X})}(\mathbf{y})}$$

and the residual

$$\mathbf{e} = \mathbf{y} - \widehat{\mathbf{y}}$$

satisfies

$$\boxed{\mathbf{X}^\top\mathbf{e} = 0.}$$

Statistical properties of least squares

Assume the linear model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \mathbb{E}[\boldsymbol{\varepsilon}] = 0, \qquad \mathrm{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}.$$

Then the least-squares estimator

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$$

is unbiased:

$$\boxed{\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}}$$

and

$$\boxed{\mathrm{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}.}$$

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Important consequence: If $\mathbf{X}^\top \mathbf{X}$ is ill-conditioned, some directions of $\hat{\boldsymbol{\beta}}$ have large variance.

$$\boxed{\text{ill-conditioning} \quad \Longrightarrow \quad \text{unstable parameter estimation.}}$$

High-dimensional linear regression

Challenges:

When the number of features d is large relative to n, or when features are strongly correlated:

- *Stability / generalization:* least squares may have large variance and become highly sensitive to the training data;
- *Identifiability:* if $d + 1 > n$, $\mathbf{X}^\top \mathbf{X}$ cannot be invertible;
- *Interpretability:* we may want to identify a small subset of relevant features.

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Regularization:

$$\boxed{F(\beta) = \frac{1}{2}\|\mathbf{y} - \mathbf{X}\beta\|^2 + \lambda R(\beta).}$$

- Ridge: $R(\beta) = \frac{1}{2}\|\beta_{1:d}\|_2^2$;
- LASSO: $R(\beta) = \|\beta_{1:d}\|_1$;
- Best subset selection: $R(\beta) = \|\beta_{1:d}\|_0$.

The intercept is left unpenalized.

Regularization changes the optimization problem

Regularizer	**Structure**	**Optimization**
$\frac{1}{2}\|\boldsymbol{\beta}_{1:d}\|_2^2$	smooth, convex	closed form / gradient methods
$\|\boldsymbol{\beta}_{1:d}\|_1$	nonsmooth, convex	proximal methods
$\|\boldsymbol{\beta}_{1:d}\|_0$	nonconvex, combinatorial	subset-selection methods

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Same data-fitting term $+$ different regularizer $\Longrightarrow$ different optimization problem.

Proximal gradient for regularized regression

Consider the regularized problem

$$\min_{\beta \in \mathbb{R}^{d+1}} F(\beta) = \underbrace{\frac{1}{2}\|\mathbf{y} - \mathbf{X}\beta\|^2}_{\Phi(\beta)} + \lambda R(\beta).$$

The data-fitting term is smooth, with

$$\nabla \Phi(\beta) = \mathbf{X}^\top(\mathbf{X}\beta - \mathbf{y}).$$

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$$\boxed{\beta^{(t+1)} = \operatorname{prox}_{\gamma\lambda R}\left(\beta^{(t)} - \gamma \nabla \Phi(\beta^{(t)})\right).}$$

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Gradient step for data fitting + proximal step for regularization.

Three penalties, three proximal updates

For $\mathbf{u} \in \mathbb{R}^{d+1}$, the proximal step is explicit and acts componentwise. The intercept is left unchanged.

- **Ridge:** $R(\beta) = \frac{1}{2}\|\beta_{1:d}\|_2^2$; for $j = 1, \ldots, d$:

$$\boxed{[\text{prox}_{\gamma\lambda R}(\mathbf{u})]_j = \frac{1}{1+\gamma\lambda}\, u_j}$$

shrinkage.

- **LASSO:** $R(\beta) = \|\beta_{1:d}\|_1$; for $j = 1, \ldots, d$:

$$\boxed{[\text{prox}_{\gamma\lambda R}(\mathbf{u})]_j = \text{sign}(u_j)(|u_j| - \gamma\lambda)_+}$$

soft thresholding.

- **Best subset:** $R(\beta) = \|\beta_{1:d}\|_0$; for $j = 1, \ldots, d$:

$$\boxed{[\text{prox}_{\gamma\lambda R}(\mathbf{u})]_j = u_j\, \mathbf{1}_{\{|u_j| > \sqrt{2\gamma\lambda}\}}}$$

hard thresholding.

Robust regression

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Approach: M-estimation

Replace the quadratic loss by a more robust potential ρ:

$$F(\boldsymbol{\beta}) = \sum_{i=1}^{n} \rho\left(y_i - \mathbf{x}_i^\top \boldsymbol{\beta}\right).$$

Typical assumptions on ρ:

- $\rho(e) \geq 0$ and $\rho(0) = 0$;
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With $r_i(\boldsymbol{\beta}) = y_i - \mathbf{x}_i^\top \boldsymbol{\beta}$, a differentiable minimizer satisfies

$$\sum_{i=1}^{n} \dot{\rho}\big(r_i(\widehat{\boldsymbol{\beta}})\big)\mathbf{x}_i = 0.$$

From robust loss to weighted least squares

Recall the robust regression criterion

$$F(\boldsymbol{\beta}) = \sum_{i=1}^{n} \rho\big(r_i(\boldsymbol{\beta})\big), \qquad r_i(\boldsymbol{\beta}) = y_i - \mathbf{x}_i^\top \boldsymbol{\beta}.$$

Assume that ρ is even, and write $\rho(r) = \phi(|r|)$.

For $u > 0$, define $\boxed{\omega(u) = \dfrac{\dot{\phi}(u)}{u}.}$ Then, for $r \neq 0$, $\boxed{\dot{\rho}(r) = \omega(|r|)\, r.}$

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If the weights were fixed, this would be exactly the normal equation of a **weighted least-squares problem**.

IRLS as a majorization-minimization method

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Using $\omega(u) = \frac{\dot{\phi}(u)}{u}$, (with $\omega(0)$ defined by continuity whenever the limit exists), we obtain

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Applied to all residuals, $r_i(\boldsymbol{\beta}) = y_i - \mathbf{x}_i^\top \boldsymbol{\beta}$, this gives a quadratic surrogate of the form

$$\boxed{Q(\boldsymbol{\beta} \mid \boldsymbol{\beta}^{(k)}) = \frac{1}{2}\sum_{i=1}^{n} \omega\left(|r_i(\boldsymbol{\beta}^{(k)})|\right) r_i(\boldsymbol{\beta})^2 + \text{constant}.}$$

Therefore each MM iteration reduces to a **weighted least-squares problem**.

Examples of functions ρ

	$\rho(x)$	$\omega(x)$ (exercise)
Convex	$\lvert x\rvert - \delta \log(\lvert x\rvert/\delta + 1)$	
	$\begin{cases} x^2 & \text{if } \lvert x\rvert < \delta \\ 2\delta\lvert x\rvert - \delta^2 & \text{otherwise} \end{cases}$	
	$\log(\cosh(x))$	
	$(1 + x^2/\delta^2)^{\kappa/2} - 1$	
Nonconvex	$1 - \exp(-x^2/(2\delta^2))$	
	$x^2/(2\delta^2 + x^2)$	
	$\begin{cases} 1 - (1 - x^2/(6\delta^2))^3 & \text{if } \lvert x\rvert \leq \sqrt{6}\delta \\ 1 & \text{otherwise} \end{cases}$	
	$\tanh(x^2/(2\delta^2))$	
	$\log(1 + x^2/\delta^2)$	

$\delta \in]0, +\infty[$, $\kappa \in [1, 2]$.

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Starting from $\boldsymbol{\beta}^{(0)}$, iterate:

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large residual $\Longrightarrow$ small weight $\Longrightarrow$ reduced influence.

Linear regression: take-home messages

- Least squares provides a simple, interpretable baseline:

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- Changing the regularizer changes the **optimization problem**.
- Changing the loss changes the **robustness properties** and may lead to algorithms such as IRLS.